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Radical Reactions

Alkene Difunctionalization Triggered by a Stabilized Allenyl Radical: Concomitant Installation of Two Unsaturated C–C Bonds

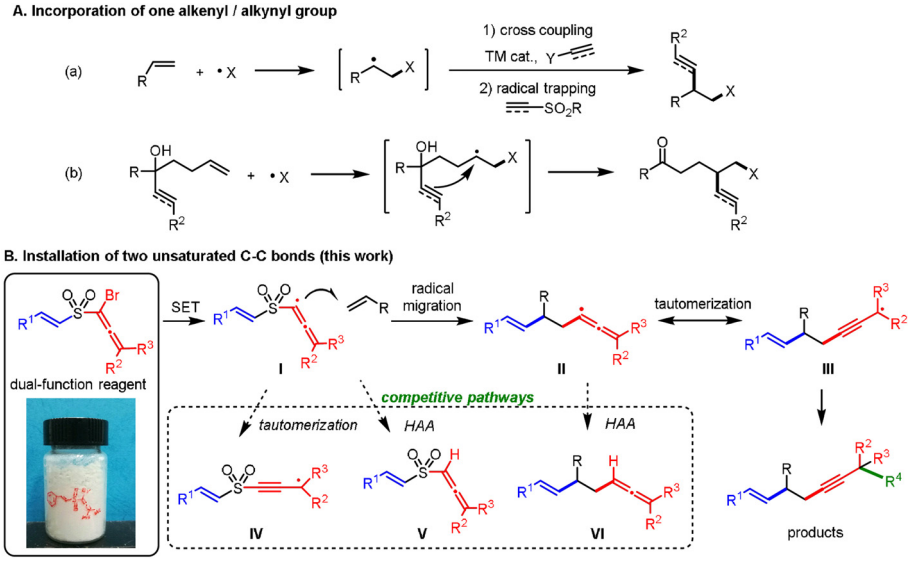
*Yunlong Wei, Hong Zhang, Xinxin Wu, and Chen Zhu**

Dedicated to the centenary of Xiamen University

Abstract: *Radical-mediated difunctionalization of alkenes provides a promising approach to introduce one alkenyl or alkynyl group to target compounds. However, simultaneous installation of two unsaturated C–C bonds via alkene difunctionalization remains elusive, attributable to the high instability and transient lifetimes of alkenyl and alkynyl radicals. Herein, we report the photocatalytic 1,2-alkynylalkenylation and 1,2-enynylalkenylation of alkenes for the first time, triggered by the intermolecular addition of a stabilized allenyl radical to an alkene. A portfolio of strategically designed, easily accessible dual-function reagents are applied to a radical docking-migration cascade. The protocol has broad substrate scope and efficiently increases the degree of unsaturation.*

Unsaturated carbon–carbon (C–C) bonds, alkenes and alkynes in particular, are ubiquitous structural units in natural products, pharmaceuticals, materials science, and also have significant applications in synthetic chemistry.[1] Over the past few decades, radical-mediated alkene difunctionalizations offers a robust toolkit to obtain numerous high value-added products.[2] This process facilitates the incorporation of an alkenyl or alkynyl into target compounds, giving rise to multi-functionalized alkene or alkyne adducts. The general protocol is initiated by the addition of an external radical to an alkene, forming a new alkyl radical intermediate which is involved in the ensuing transition-metal catalyzed cross-couplings,[2d,e,k,n,o] or trapped by radical acceptors,[2g,3] such as the Fuchs reagent (Scheme 1 A, a). Alternatively, the nascent alkyl radical can undergo an intramolecular alkenyl or alkynyl migration by harnessing the tertiary allylic/propargylic alcohols as specific substrates (Scheme 1 A, b).[4] Nevertheless, these approaches are restricted to mounting a single alkenyl or alkynyl group at alkenes.

Simultaneous installation of two unsaturated C–C bonds at both sides of an alkene, which efficiently increases the unsaturation degree and complexity of molecules, is highly challenging. The main obstacle is inducing typically short-lived alkenyl or alkynyl radical to add intermolecularly to alkenes. Of note, alkynyl radicals are normally only found in low density atmospheric and interstellar environments.[5] In addition, the strong tendency of alkenyl radicals towards hydrogen atom abstraction (HAA) from any available source also competes with the desired radical addition.[6] As a result, despite the great progress made in radical-mediated difunctionalization of alkenes, concomitant installation of two unsaturated C–C bonds by alkene difunctionalization still remains unmet.



Scheme 1. Incorporation of unsaturated C–C bonds via radical-mediated alkene difunctionalization.

Prompted by the applications of functional group migration in many elusive alkene difunctionalization reactions,[7] we

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conceive of designing a novel, practical dual-function reagent for the 1,2-alkynylalkenylation (Scheme 1B). These reagents are easily accessed within a few steps in gram-scale, in which an alkenyl and a bromoallenyl group are tethered by a sulfone. In the hypothesis, the allenyl radical **I** arising from the dual-function reagent is supposed to be stabilized by the sulfonyl, thus enabling its addition to an alkene. After a radical migration cascade, a new allenyl radical **II** is generated which might tautomerize to the propargyl radical **III** without the stabilization effect of the sulfonyl group. The subsequent functionalization of **III** furnishes the desired 1,2-alkynylalkenylation product. However, there are still many concerns. Free allenyl radicals are generally detected by ESR below 100 K,[8] and mostly involved in transition metal-catalyzed cross couplings,[9] intramolecular cyclization,[10] and hydrogen/halogen abstraction.[11] There is no evidence that the allenyl radical could be stabilized by a sulfonyl group. Therefore, the intermolecular addition of transient allenyl radical to alkene in this hypothesis is still questionable. The tautomerization between allenyl radical and propargyl radical is also a vital yet uncertain issue to the transformation.[12] If the competitive conversion of **I** to **IV** occurs or the anticipated conversion of **II** to **III** does not occur, the transformation will be terminated. Moreover, the possible HAA process induced by the sp^2 allenyl radical **I** or **II** would also inhibit the reaction.

Herein, we disclose the proof-of-principle studies for the radical-mediated 1,2-alkynylalkenylation and 1,2-enynylalkenylation of alkenes. The sulfone in the dual-function reagent not only acts as a linker, but the stabilizing group for the allenyl radical which enables the intermolecular addition step. To the best of our knowledge, this is the first example of concomitant installation of two unsaturated C–C bonds across alkenes.

The reaction of styrene with dual-function reagent **1a** was implemented to optimize reaction conditions. After a systematic survey (for details, see SI), the desired 1,5-enyne product **3a** was obtained with 72% yield under the photochemical conditions by using acetonitrile and methanol as mixed solvents (Figure 1). The ratio (3:1) of methanol to acetonitrile

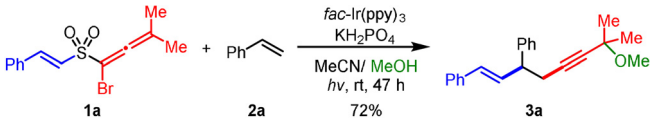


Figure 1. Optimized conditions for 1,2-alkynylalkenylation of alkenes.

was crucial to the reaction outcome. Insufficient methanol resulted in a lower yield of **3a**, while use of methanol as the sole solvent entirely impeded the transformation. In addition, other bases such as Na_2CO_3, K_2CO_3, Et_3N, and DIPEA were not effective for the reaction.

With the optimized reaction conditions in hand, we set about assessing the generality of the protocol (Figure 2). Firstly, a variety of aromatic alkenes bearing diverse functional groups were tested. The electronic properties and positional change (*para*-, *meta*-, or *ortho*-) of substituents did not have much impact on the reaction outcomes. Bromide (**3d** and **3n**) and pinacol boronate (**3i**) remained intact in the

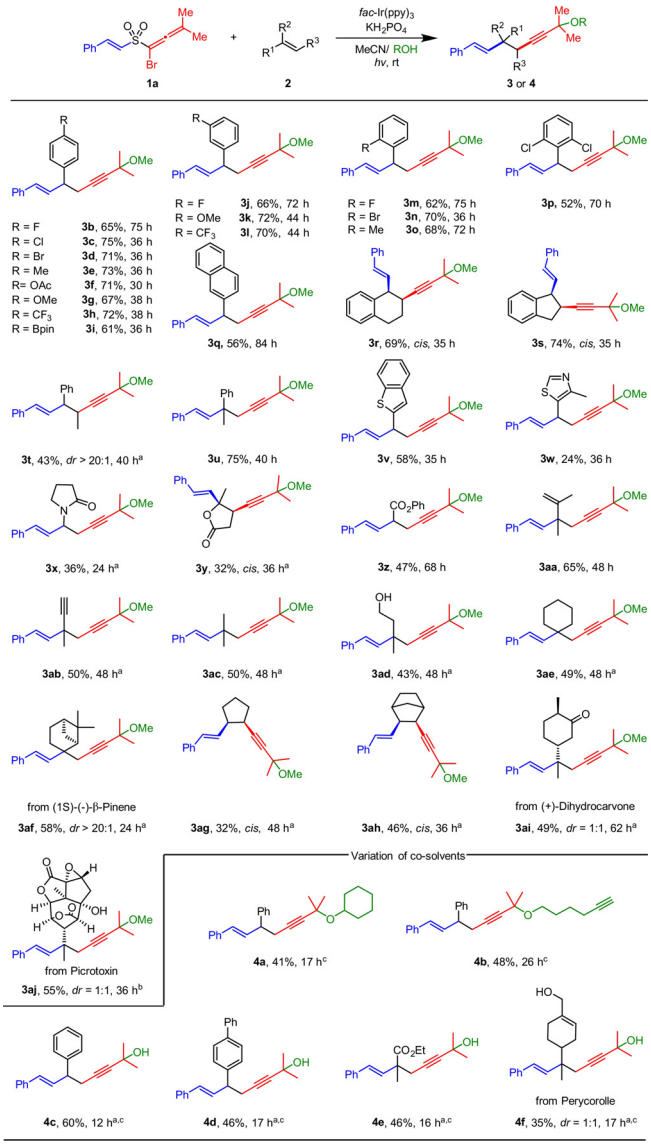


Figure 2. Scope of alkenes. Reaction conditions: **1a** (0.2 mmol), **2** (1.5 equiv), *fac*-Ir(ppy)$_3$ (1 mol%), and KH_2PO_4 (2 equiv) in MeCN/MeOH (0.5/ 1.5 mL) under N_2 at rt, irradiated with 5 W blue LEDs. [a] **2** (3 equiv) and *fac*-Ir(ppy)$_3$ (2 mol%). [b] **2** (2 equiv) and *fac*-Ir(ppy)$_3$ (2 mol%). [c] MeCN/ ROH or MeCN/ H_2O (1.5/ 0.3 mL).

reaction, providing a platform for further product elaboration. Internal alkenes, normally regarded as unfavorable substrates in radical difunctionalization, presented comparable reactivity herein. Notably, products **3r** and **3s** were obtained with unique *cis*-stereoselectivity, attributed to the formation of five-membered cyclic transition state during the alkenyl migration process. Heteroaryl-substituted alkenes were also compatible with the protocol (**3v** and **3w**). Functionalized alkenes such as enamide, β,γ-butenolide, and phenyl acrylate were suitable to this protocol and afforded the corresponding products in useful yields (**3x**–**3z**), but the reaction with vinyl ether only resulted in a very low yield (< 10%). 1,3-Diene and 1,3-enyne served as suitable precursors, leading to valuable products with high degrees of unsaturation (**3aa** and **3ab**). In the latter case, the reaction

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regioselectively occurred at the alkenyl part despite the presence of an alkyne which is also sensitive to radical transformations. Remarkably, difunctionalization of aliphatic alkenes including low-boiling-point representatives, such as isobutene, proceeded smoothly, in which a set of all-carbon quaternary centers were constructed regardless of steric congestion (**3ac**–**3ae**). Intriguingly, the reaction of β-pinene, a widely used radical-clock probe, gave the unusual 1,2-difunctionalized product **3af** instead of common ring-opening adduct, indicating that the intramolecular radical 5-exo-trig cyclization is faster than the competitive ring-opening process. However, performing the radical-clock experiments with more reactive vinylcyclopropanes (VCPs, ≈10 times to β-pinene)[13] led to either the ring-opening product or the difunctionalized product dependent on the electronic property of VCPs (for details, see SI), suggesting that the rate of 5-exo-trig cyclization may be comparable to the ring-opening rates of VCPs. Similar to **3r** and **3s**, the reaction of cyclopentene and norbornene also exclusively delivered the *cis*-adducts **3ag** and **3ah**. The method could be applied to the late-stage modification of complex natural products. For instance, **3ai** and the highly functionalized **3aj** were readily obtained from dihydrocarvone and picrotoxin, respectively, manifesting that this protocol is of broad functional group tolerance.

It was anticipated that methanol could be replaced with other co-solvents. Indeed, cyclohexanol and 5-hexynyl-1-ol also proved suitable and afforded the corresponding products with acceptable yields (**4a** and **4b**). Terminal alkynes were tolerated in the reaction. Notably, synthetically valuable propargylic alcohols **4c**–**4f** were obtained by using water as co-solvent.

To further extend the product diversity, a portfolio of dual-function reagents was prepared and subjected to the reaction with styrene (Figure 3). The two methyl groups on allene could be varied to a cycloalkyl or heterocycle, leading to the corresponding adducts with slightly lower yields (**5a**–**5c**). Altering a methyl to a larger aliphatic group in the reagent also presented a comparable yield (**5d**). Replacing methyl by phenyl appreciably decreased the yields of products (**5e** and **5f**), probably due to the negative p-π conjugate effect from aryl group which suppressed free tautomerization between the allenyl and propargylic radical intermediates. Various substitutions on the alkenyl part were also apt to afford the desired products regardless of steric and electronic effects (**5g**–**5k**). However, sharply increasing the steric hindrance of the dual-function reagent (e.g. **1m**) would impede the formation of five-membered cyclic transition state as well as the ensuing alkenyl migration, thus resulting in the low yield of **5l**. Remarkably, the use of dual-function reagent **1n** containing a *Z,E*-diene moiety simultaneously incorporated an alkynyl and an *E,E*-diene into product **5m**, indicating a superb stereocontrolled alkenylation process.

1,3-Conjugated enynes represent a class of highly transformable building blocks in synthetic chemistry.[14] Inspired by the 1,2-alkynylalkenylation pathway, we envisioned that, in the absence of nucleophilic co-solvent, the transformation might give rise to 1,3-enynes and lead to 1,2-enynylalkenylation of alkenes through an alternative deprotonation.

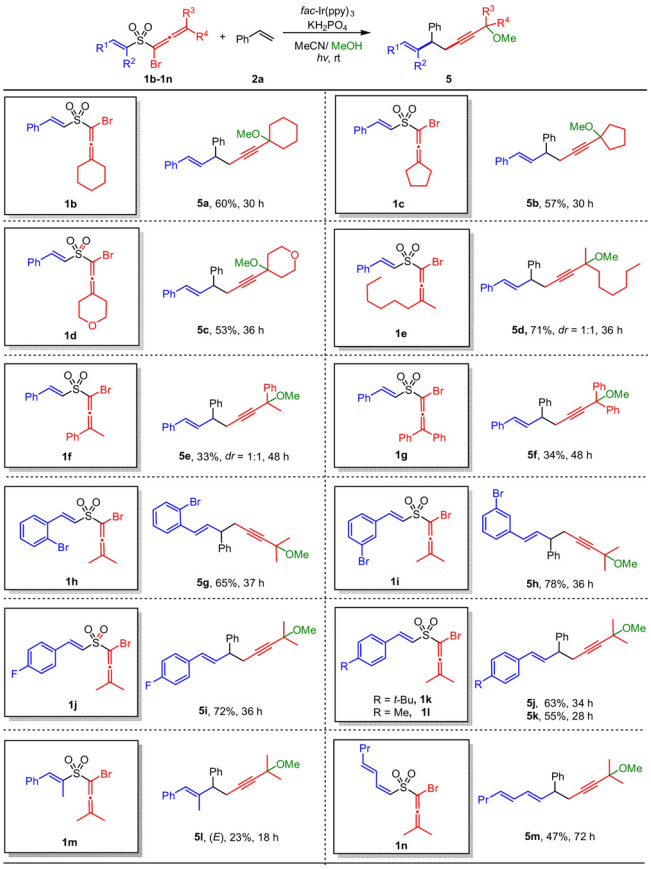


Figure 3. Assessment of dual-function reagents. Reaction conditions: **1b**–**1n** (0.2 mmol), **2a** (1.5 equiv), *fac*-Ir(ppy)$_3$ (1 mol%), and KH_2PO_4 (2 equiv) in MeCN/ MeOH (0.5/ 1.5 mL) under N_2 at rt, irradiated with 5 W blue LEDs.

Indeed, the 1,2-enynylalkenylation occurred as expected under slightly modified conditions by using DMF as sole solvent. Representative examples were listed to illustrate the generality of the protocol (Figure 4). Various substituents on

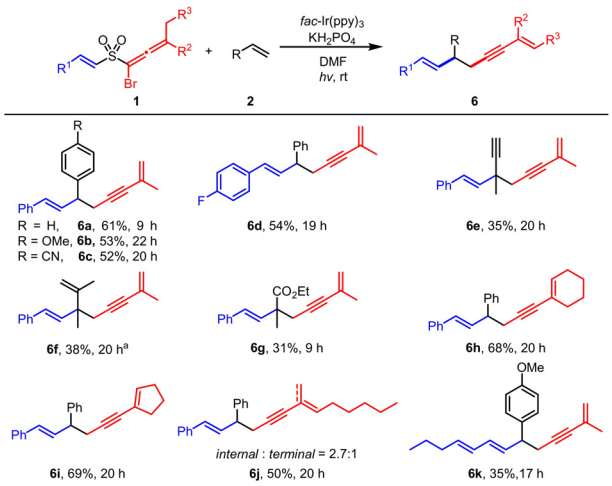
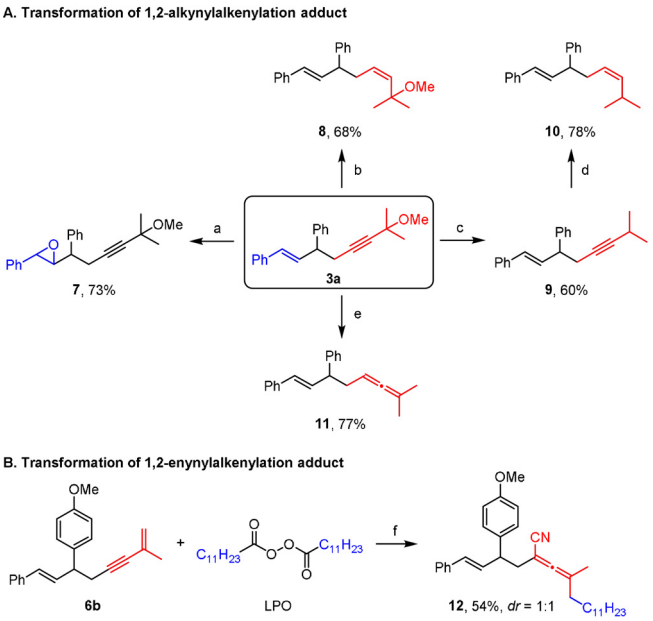


Figure 4. Investigation of 1,2-enynylalkenylation of alkenes. Reaction conditions: **1** (0.2 mmol), **2** (3 equiv), *fac*-Ir(ppy)$_3$ (2 mol%), and KH_2PO_4 (1 equiv) in DMF (2 mL) under N_2 at rt, irradiated with 30 W blue LEDs. [a] **2** (1.5 equiv).

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the alkene or dual-function reagent were well tolerated, leading to the corresponding products (**6a**–**6d**). Notably, difunctionalization of 1,3-enyne and 1,3-diene furnished the intricately branched adducts with dense unsaturated groups that are otherwise hard to obtain (**6e** and **6f**). The undesired over-functionalization of the residual terminal alkenyl in products were clearly observed, which accounted for the lower yields. The conversion with highly electron-deficient acrylate also proceeded (**6g**), albeit in modest yield. Utilization of reagent **1b** and **1c** afforded the internal enyne products in synthetically useful yields (**6h** and **6i**), as the over-functionalization of products without terminal enynyl was entirely avoided. The reaction with reagent **1e** resulted in a mixture of internal and terminal enynyl products (**6j**). A linear multi-unsaturated molecule **6k** was also readily synthesized with the use of reagent **1n**.

The synthetic value of the method was showcased by transformation of the products into other valuable molecules (Scheme 2). In the presence of *m*-CPBA, **3a** was converted to epoxide **7**, in which the alkynyl moiety remained intact. Selective Lindlar reduction of **3a** gave rise to *Z,E*-diene **8**, providing a formal protocol for radical dialkenylation of alkenes. The propargylic methoxy in **3a** could be easily removed, and the resultant product **9** was stereoselectively reduced by DIBAL-H to give *Z,E*-diene **10**. Interestingly, the propargylic methoxy group could also function as a leaving group, leading to the formal allenylalkenylation adduct **11** with the use of DIBAL-H. Moreover, the transformation of enyne **6b** provided another formal 1,2-allenylalkenylation approach, furnishing the fully substituted allene **12** via the copper-catalyzed radical cyanoalkylation.



Scheme 2. Product transformations. Conditions a: *m*-CPBA (3 equiv), $NaHCO_3$ (3 equiv), DCM/H_2O, 20 h. Conditions b: Lindlar's catalyst (5 wt%, 20 mol%), quinoline (0.5 equiv), 1 atm H_2, dry EtOAc, rt, 15 min. Conditions c: Cp_2ZrCl_2 (1.6 equiv), *n*-BuLi (2.5 M, 3.2 equiv), THF, −78 °C for 1 h, then rt for 3 h. Conditions d: DIBAL-H (2 equiv), toluene, 50 °C, 12 h. Conditions e: DIBAL-H (4 equiv), toluene, 80 °C, 21 h. Conditions f: $Cu(OAc)_2$ (5 mol%), 1,10-phenanthroline (7 mol%), LPO (3 equiv), TMSCN (4 equiv), dry DCM, 50 °C, 42 h.

Subsequent efforts were focused on the reaction mechanism investigation. Firstly, the Stern–Volmer study showed that the excited-state of $Ir(ppy)_3$ was readily oxidatively quenched by **1a** (Figure S1). This was also verified by the reduction potential of **1a**. The value ($E_{p/2} = -1.55$ V vs. SCE) illustrated that the C–Br bond of **1a** could be reduced by Ir^{III}* ($E_{1/2}^{IV/III*} = -1.73$ V vs. SCE) (Figure S2).[15] The quantum yield of the reaction between **1a** and styrene was 0.14 (Figure S3), suggesting that a photocatalytic cycle might be involved as the main pathway. The light on-off experiments also supported this conclusion that the reaction outcome was not increased without the light irradiation (Figure S4).

A plausible mechanism is depicted in Figure 5. Single-electron transfer from the excited-state of $Ir(ppy)_3$ to dual-function reagent **1a** affords the allenyl radical **I**, which is verified by radical trap experiment with TEMPO (for details, see SI), and Ir^{IV} species. This sulfone-stabilized allenyl radical **I** is efficiently trapped by alkene to generate the intermediate **II**. The ensuing cascade of intramolecular alkenyl migration and SO_2 extrusion results in a new allenyl radical **IV**. The tautomerization of **IV** to a more stable propargylic radical **V** followed by single-electron oxidation with the Ir^{IV} species gives rise to the cation intermediate **VI**, and regenerates the ground-state Ir^{III} catalyst. The reaction of **VI** with nucleophilic co-solvents (e.g. ROH, H_2O, etc.) furnishes the adduct **3**, **4** and **5**. Alternatively, in the absence of nucleophiles the reaction leads to the enyne **6** via the deprotonation of **VI**.

In summary, we have disclosed the first example of concomitant incorporation of two unsaturated C–C bonds into alkenes by the photocatalytic 1,2-alkynylalkenylation and 1,2-enynylalkenylation of alkenes. A portfolio of strategically designed dual-function reagents are applied to the transformation. The sulfone in the dual-function reagent

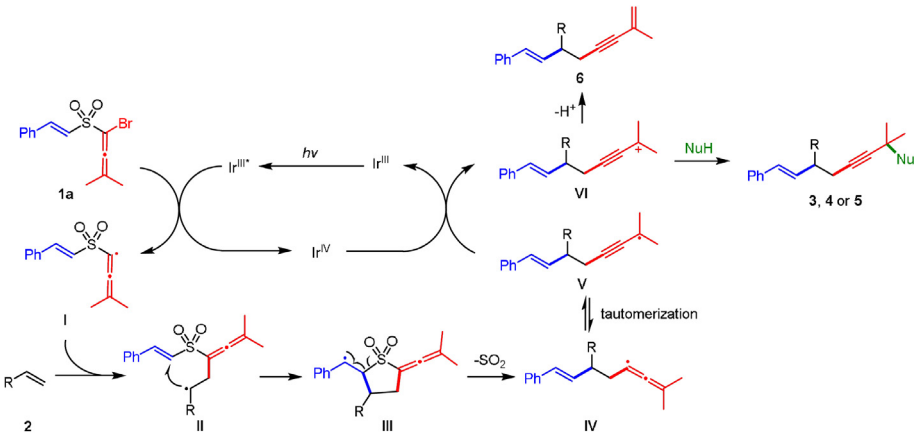


Figure 5. Proposed reaction mechanism.

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stabilizes the highly active allenyl radical intermediate, thus enabling the elusively intermolecular addition of allenyl radical to alkene. The reaction features a broad scope of alkenes that both activated and unactivated alkenes are suitable substrates. Moreover, the protocol is of great synthetic value, affording a diversity of complex unsaturated molecules with high unsaturation degree, which are otherwise difficult to prepare. This ingenious application of docking-migration strategy opens up a new vista for radical difunctionalization of alkenes.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: alkene difunctionalization · alkyne · allenyl radical · radical reactions · rearrangement

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